

0902.0431: residuals

equations measured 2026-09-15T14:00:51Z against the model of 2026-09-15

Corrected (4)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_EQ0756 (was)	125	■ 0.119	$\begin{array}{l} \backslash begin{array}{llllllllll} \backslash lambda_{\{0\}}-\backslash lambda_{\{1\}}=1 \\ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \backslash lambda_{\{0\}}+\backslash lambda_{\{1\}}=1 \ \& \\ 2 \ \& \ 2 \ \& \ 2 \ \& \ 2 \ \& \ 1 \ \backslash \backslash \backslash lambda_{\{0\}}-\backslash lambda_{\{2\}}=1 \ \& \ 1 \ \& \\ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \backslash lambda_{\{0\}}+\backslash lambda_{\{2\}}=1 \ \& \ 1 \ \& \ 2 \\ \& \ 2 \ \& \ 2 \ \& \ 1 \ \backslash \backslash \backslash lambda_{\{0\}}-\backslash lambda_{\{3\}}=1 \ \& \ 1 \ \& \ 1 \ \& \ 0 \\ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \backslash lambda_{\{0\}}+\backslash lambda_{\{3\}}=1 \ \& \ 1 \ \& \ 1 \ \& \ 2 \ \& \\ 2 \ \& \ 1 \ \backslash \backslash \backslash lambda_{\{1\}}-\backslash lambda_{\{2\}}=0 \ \& \ 1 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \\ 0 \ \& \ 0 \ \backslash lambda_{\{1\}}+\backslash lambda_{\{2\}}=0 \ \& \ 1 \ \& \ 2 \ \& \ 2 \ \& \ 2 \ \& \ 1 \\ \backslash \backslash \backslash lambda_{\{1\}}-\backslash lambda_{\{3\}}=0 \ \& \ 1 \ \& \ 1 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \\ \& \ \backslash lambda_{\{1\}}+\backslash lambda_{\{3\}}=0 \ \& \ 1 \ \& \ 1 \ \& \ 2 \ \& \ 2 \ \& \ 1 \ \backslash \backslash \\ \backslash lambda_{\{2\}}-\backslash lambda_{\{3\}}=0 \ \& \ 0 \ \& \ 1 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \\ \backslash lambda_{\{2\}}+\backslash lambda_{\{3\}}=0 \ \& \ 0 \ \& \ 1 \ \& \ 2 \ \& \ 2 \ \& \ 1 \ \backslash \backslash \ \& \ \& \\ \backslash lambda_{\{0\}}+\frac{1}{2}\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=1 \ \& \ 1 \\ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ \& \ \backslash \backslash \ \& \ \& \ \backslash lambda_{\{1\}}+\frac{1}{2}\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=0 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \\ 1 \ \& \ \& \ \backslash \backslash \ \& \ \& \ \backslash lambda_{\{2\}}+\frac{1}{2}\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=0 \ \& \ 0 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ \& \ \backslash \backslash \ \& \ \& \ + \\ \frac{1}{2}\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=0 \ \& \ 0 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ \& \ \backslash \backslash \ \& \ \& \ + \\ \frac{1}{2}\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=0 \ \& \ 0 \ \& \ 0 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \ \& \ \end{array}$	$\begin{array}{llllll} \lambda_0-\lambda_1=1 & 0 & 0 & 0 & 0 & 0 \\ \lambda_0-\lambda_2=1 & 1 & 0 & 0 & 0 & 0 \\ \lambda_0-\lambda_3=1 & 1 & 1 & 0 & 0 & 0 \\ \lambda_1-\lambda_2=0 & 1 & 0 & 0 & 0 & 0 \\ \lambda_1-\lambda_3=0 & 1 & 1 & 0 & 0 & 0 \\ \lambda_2-\lambda_3=0 & 0 & 1 & 0 & 0 & 0 \end{array} \quad \begin{array}{llllll} 0 & 0 & 0 & 0 & \lambda_0+\lambda_1=1 & 2 \\ 0 & 0 & 0 & 0 & \lambda_0+\lambda_2=1 & 1 \\ 0 & 0 & 0 & 0 & \lambda_0+\lambda_3=1 & 1 \\ 0 & 0 & 0 & 0 & \lambda_1+\lambda_2=0 & 1 \\ 0 & 0 & 0 & 0 & \lambda_1+\lambda_3=0 & 1 \\ 0 & 0 & 0 & 0 & \lambda_2+\lambda_3=0 & 0 \end{array} \quad \begin{array}{llllll} 2 & 2 & 2 & 2 & 1 & 1 \\ 2 & 2 & 2 & 2 & 1 & 1 \\ 2 & 2 & 2 & 2 & 1 & 1 \\ 2 & 2 & 2 & 2 & 1 & 1 \\ 2 & 2 & 2 & 2 & 1 & 1 \\ 1 & 2 & 2 & 2 & 1 & 1 \end{array}$ $\begin{array}{llllll} \lambda_0+\frac{1}{2}(\mu_2-\mu_3)=1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_1+\frac{1}{2}(\mu_2-\mu_3)=0 & 1 & 1 & 1 & 1 & 1 \\ \lambda_2+\frac{1}{2}(\mu_2-\mu_3)=0 & 0 & 1 & 1 & 1 & 1 \\ \lambda_3+\frac{1}{2}(\mu_2-\mu_3)=0 & 0 & 0 & 1 & 1 & 1 \end{array}$	$\begin{array}{llllll} \lambda_0-\lambda_1=1 & 0 & 0 & 0 & 0 & 0 \\ \lambda_0-\lambda_2=1 & 1 & 0 & 0 & 0 & 0 \\ \lambda_0-\lambda_3=1 & 1 & 1 & 0 & 0 & 0 \\ \lambda_1-\lambda_2=0 & 1 & 0 & 0 & 0 & 0 \\ \lambda_1-\lambda_3=0 & 1 & 1 & 0 & 0 & 0 \\ \lambda_2-\lambda_3=0 & 0 & 1 & 0 & 0 & 0 \end{array} \quad \begin{array}{llllll} \lambda_0+\lambda_1=1 & 2 & 2 & 2 & 2 & 1 \\ \lambda_0+\lambda_2=1 & 1 & 2 & 2 & 2 & 1 \\ \lambda_0+\lambda_3=1 & 1 & 1 & 2 & 2 & 1 \\ \lambda_1+\lambda_2=0 & 1 & 2 & 2 & 2 & 1 \\ \lambda_1+\lambda_3=0 & 1 & 1 & 2 & 2 & 1 \\ \lambda_2+\lambda_3=0 & 0 & 1 & 2 & 2 & 1 \end{array}$ $\begin{array}{llllll} \lambda_0+\frac{1}{2}(\mu_2-\mu_3)=1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_1+\frac{1}{2}(\mu_2-\mu_3)=0 & 1 & 1 & 1 & 1 & 1 \\ \lambda_2+\frac{1}{2}(\mu_2-\mu_3)=0 & 0 & 1 & 1 & 1 & 1 \\ +\frac{1}{2}(\mu_2-\mu_3)=0 & 0 & 0 & 1 & 1 & 1 \end{array}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E00756 (now)	125	—	$\begin{array}{l} \backslash begin{array}{llllllllllll} \lambda_0 - \lambda_1 = 1 \\ \& 0 \& 0 \& 0 \& 0 \& 0 \& 0 \& 0 \& \lambda_0 + \lambda_1 = 1 \\ 2 \& 2 \& 2 \& 2 \& 2 \& 1 \backslash \backslash \lambda_0 - \lambda_2 = 1 \& 1 \& \\ 0 \& 0 \& 0 \& 0 \& 0 \& 0 \& \lambda_0 + \lambda_2 = 1 \& 1 \& 2 \\ \& 2 \& 2 \& 1 \backslash \backslash \lambda_0 - \lambda_3 = 1 \& 1 \& 1 \& 0 \\ \& 0 \& 0 \& 0 \& 0 \& \lambda_0 + \lambda_3 = 1 \& 1 \& 1 \& 2 \& \\ 2 \& 1 \backslash \backslash \lambda_1 - \lambda_2 = 0 \& 1 \& 0 \& 0 \& 0 \& 0 \& \\ 0 \& 0 \& \lambda_1 + \lambda_2 = 0 \& 1 \& 2 \& 2 \& 2 \& 1 \\ \backslash \backslash \lambda_1 - \lambda_3 = 0 \& 1 \& 1 \& 0 \& 0 \& 0 \& 0 \& \\ \& \lambda_1 + \lambda_3 = 0 \& 1 \& 1 \& 2 \& 2 \& 1 \backslash \backslash \\ \lambda_2 - \lambda_3 = 0 \& 0 \& 1 \& 0 \& 0 \& 0 \& 0 \& \\ \lambda_2 + \lambda_3 = 0 \& 0 \& 1 \& 2 \& 2 \& 1 \backslash \backslash \& \& \\ \lambda_0 + \frac{1}{2}(\mu_2 - \mu_3) = 1 \& 1 \& 1 \& 1 \& 1 \& 1 \& \\ \& 1 \& 1 \& 1 \& 1 \& 1 \& 1 \& \& \& \backslash \backslash \& \& \lambda_1 + \\ \frac{1}{2}(\mu_2 - \mu_3) = 0 \& 1 \& 1 \& 1 \& 1 \& \\ \& 1 \& 1 \& 1 \& \& \& \backslash \backslash \& \& \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \& 0 \& 1 \& 1 \& 1 \& 1 \& \& \\ \& \backslash \backslash \& \& \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \& 0 \& 0 \& 1 \& 1 \& 1 \& \& \& \end{array}$	$\begin{array}{ll} \lambda_0 - \lambda_1 = 1 & 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \\ \lambda_0 - \lambda_2 = 1 & 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \\ \lambda_0 - \lambda_3 = 1 & 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \\ \lambda_1 - \lambda_2 = 0 & 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \\ \lambda_1 - \lambda_3 = 0 & 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \\ \lambda_2 - \lambda_3 = 0 & 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \end{array} \quad \begin{array}{ll} \lambda_0 + \lambda_1 = 1 & 2 \ 2 \ 2 \ 2 \ 1 \\ \lambda_0 + \lambda_2 = 1 & 1 \ 2 \ 2 \ 2 \ 1 \\ \lambda_0 + \lambda_3 = 1 & 1 \ 1 \ 2 \ 2 \ 1 \\ \lambda_1 + \lambda_2 = 0 & 1 \ 2 \ 2 \ 2 \ 1 \\ \lambda_1 + \lambda_3 = 0 & 1 \ 1 \ 2 \ 2 \ 1 \\ \lambda_2 + \lambda_3 = 0 & 0 \ 1 \ 2 \ 2 \ 1 \end{array}$ $\begin{array}{ll} \lambda_0 + \frac{1}{2}(\mu_2 - \mu_3) = 1 & 1 \ 1 \ 1 \ 1 \ 1 \ 1 \\ \lambda_1 + \frac{1}{2}(\mu_2 - \mu_3) = 0 & 1 \ 1 \ 1 \ 1 \ 1 \\ \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) = 0 & 0 \ 1 \ 1 \ 1 \ 1 \ 1 \\ \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3) = 0 & 0 \ 0 \ 1 \ 1 \ 1 \ 1 \end{array}$	$\begin{array}{ll} \lambda_0 - \lambda_1 = 1 & 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \\ \lambda_0 - \lambda_2 = 1 & 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \\ \lambda_0 - \lambda_3 = 1 & 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \\ \lambda_1 - \lambda_2 = 0 & 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \\ \lambda_1 - \lambda_3 = 0 & 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \\ \lambda_2 - \lambda_3 = 0 & 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \end{array} \quad \begin{array}{ll} \lambda_0 + \lambda_1 = 1 & 2 \ 2 \ 2 \ 2 \ 1 \\ \lambda_0 + \lambda_2 = 1 & 1 \ 2 \ 2 \ 2 \ 1 \\ \lambda_0 + \lambda_3 = 1 & 1 \ 1 \ 2 \ 2 \ 1 \\ \lambda_1 + \lambda_2 = 0 & 1 \ 2 \ 2 \ 2 \ 1 \\ \lambda_1 + \lambda_3 = 0 & 1 \ 1 \ 2 \ 2 \ 1 \\ \lambda_2 + \lambda_3 = 0 & 0 \ 1 \ 2 \ 2 \ 1 \end{array}$ $\begin{array}{ll} \lambda_0 + \frac{1}{2}(\mu_2 - \mu_3) = 1 & 1 \ 1 \ 1 \ 1 \ 1 \ 1 \\ \lambda_1 + \frac{1}{2}(\mu_2 - \mu_3) = 0 & 1 \ 1 \ 1 \ 1 \ 1 \\ \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) = 0 & 0 \ 1 \ 1 \ 1 \ 1 \ 1 \\ \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3) = 0 & 0 \ 0 \ 1 \ 1 \ 1 \ 1 \end{array}$
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E00922 (was)	153	■ 0.102	$f^{-1}\left(\sum_{i<j<k}x_{ij\,k}\,\boldsymbol{e}_{ij}\wedge\boldsymbol{e}_{jk}\right)=\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\overline{x}_{256}) & h(x_{145},\overline{x}_{356}) \\ h(x_{256},\overline{x}_{164}) & h(x_{264}) & h(x_{245},\overline{x}_{364}) \\ h(x_{356},\overline{x}_{145}) & h(x_{364},\overline{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431},\overline{x}_{523}) & h(x_{412},\overline{x}_{623}) \\ h(x_{523},\overline{x}_{431}) & h(x_{531}) & h(x_{512},\overline{x}_{631}) \\ h(x_{623},\overline{x}_{412}) & h(x_{631},\overline{x}_{512}) & h(x_{612})\end{array}\right),$	$f^{-1}\left(\sum_{i<j<k}x_{ijk}\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k\right)=\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\overline{x}_{256}) & h(x_{145},\overline{x}_{356}) \\ h(x_{256},\overline{x}_{164}) & h(x_{264}) & h(x_{245},\overline{x}_{364}) \\ h(x_{356},\overline{x}_{145}) & h(x_{364},\overline{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431},\overline{x}_{523}) & h(x_{412},\overline{x}_{623}) \\ h(x_{523},\overline{x}_{431}) & h(x_{531}) & h(x_{512},\overline{x}_{631}) \\ h(x_{623},\overline{x}_{412}) & h(x_{631},\overline{x}_{512}) & h(x_{612})\end{array}\right),$	$f^{-1}\left(\sum_{i<j<k}x_{ijk}\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k\right)=\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\overline{x}_{256}) & h(x_{145},\overline{x}_{356}) \\ h(x_{256},\overline{x}_{164}) & h(x_{264}) & h(x_{245},\overline{x}_{364}) \\ h(x_{356},\overline{x}_{145}) & h(x_{364},\overline{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431},\overline{x}_{523}) & h(x_{412},\overline{x}_{623}) \\ h(x_{523},\overline{x}_{431}) & h(x_{531}) & h(x_{512},\overline{x}_{631}) \\ h(x_{623},\overline{x}_{412}) & h(x_{631},\overline{x}_{512}) & h(x_{612})\end{array}\right),$
0902.0431_E00922 (now)	153	—	$f^{-1}\left(\sum_{i<j<k}x_{ij\,k}\,\boldsymbol{e}_{ij}\wedge\boldsymbol{e}_{jk}\right)=\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\overline{x}_{256}) & h(x_{145},\overline{x}_{356}) \\ h(x_{256},\overline{x}_{164}) & h(x_{264}) & h(x_{245},\overline{x}_{364}) \\ h(x_{356},\overline{x}_{145}) & h(x_{364},\overline{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431},\overline{x}_{523}) & h(x_{412},\overline{x}_{623}) \\ h(x_{523},\overline{x}_{431}) & h(x_{531}) & h(x_{512},\overline{x}_{631}) \\ h(x_{623},\overline{x}_{412}) & h(x_{631},\overline{x}_{512}) & h(x_{612})\end{array}\right),$	$f^{-1}\left(\sum_{i<j<k}x_{ijk}\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k\right)=\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\overline{x}_{256}) & h(x_{145},\overline{x}_{356}) \\ h(x_{256},\overline{x}_{164}) & h(x_{264}) & h(x_{245},\overline{x}_{364}) \\ h(x_{356},\overline{x}_{145}) & h(x_{364},\overline{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431},\overline{x}_{523}) & h(x_{412},\overline{x}_{623}) \\ h(x_{523},\overline{x}_{431}) & h(x_{531}) & h(x_{512},\overline{x}_{631}) \\ h(x_{623},\overline{x}_{412}) & h(x_{631},\overline{x}_{512}) & h(x_{612})\end{array}\right),$	$f^{-1}\left(\sum_{i<j<k}x_{ijk}\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k\right)=\left(\begin{array}{ccc}h(x_{156}) & h(x_{164},\overline{x}_{256}) & h(x_{145},\overline{x}_{356}) \\ h(x_{256},\overline{x}_{164}) & h(x_{264}) & h(x_{245},\overline{x}_{364}) \\ h(x_{356},\overline{x}_{145}) & h(x_{364},\overline{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431},\overline{x}_{523}) & h(x_{412},\overline{x}_{623}) \\ h(x_{523},\overline{x}_{431}) & h(x_{531}) & h(x_{512},\overline{x}_{631}) \\ h(x_{623},\overline{x}_{412}) & h(x_{631},\overline{x}_{512}) & h(x_{612})\end{array}\right),$
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E01032 (was)	173	■ 0.022	$\begin{aligned} & \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 2 \quad 1 \quad 1 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2+\lambda_3\right)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 1 \quad 1 \quad 1 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 3 \quad 2 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2-\lambda_3\right)+\frac{1}{2}\mu_2-\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right)-\frac{1}{2}\mu_2+\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 2 \quad 2 \quad 2 \quad 1 \quad 2 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2+\lambda_3\right)-\frac{1}{2}\mu_2+\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 2 \quad 2 \quad 1 \quad 1 \quad 2 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right)-\frac{1}{2}\mu_2-\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 2 \quad 2 \quad 1 \quad 0 \quad 2 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2-\lambda_3\right)-\frac{1}{2}\mu_2+\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right)+\frac{1}{2}\mu_3-\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 3 \quad 3 \quad 2 \quad 1 \quad 1 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2-\lambda_3\right)+\frac{1}{2}\mu_3-\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right)+\frac{1}{2}\mu_3-\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2+\lambda_3\right)+\frac{1}{2}\mu_3-\frac{2}{3}\nu=1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right)-\frac{1}{2}\mu_3+\frac{2}{3}\nu=1 \quad 3 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 3 \\ & \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2-\lambda_3\right)-\frac{1}{2}\mu_3+\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 2 \quad 1 \quad 1 \quad 1 \quad 2 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right)-\frac{1}{2}\mu_3+\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 2 \quad 1 \quad 1 \quad 0 \quad 2 \\ & \frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2+\lambda_3\right)-\frac{1}{2}\mu_3+\frac{2}{3}\nu=1 \quad 2 \quad 3 \quad 2 \quad 1 \quad 0 \quad 0 \quad 2 \end{aligned}$	(not rendered)	

Conf. MathPix confidence:

■ ≥0.9

■ 0.5-0.9

■ <0.5

— no value

LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E01032 (now)	173	—	$\begin{aligned} \kappa\pi H &= \tfrac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_0 + \tfrac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &\quad + \tfrac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \tfrac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3, \end{aligned}$	$\kappa\pi H = \tfrac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \tfrac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 + \tfrac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \tfrac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3,$	$\begin{aligned} \tfrac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \tfrac{1}{2}\mu_2 - \tfrac{2}{3}\nu &= 1 & 2 & 3 & 3 & 2 & 2 & 1 & 1 \\ \tfrac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \tfrac{1}{2}\mu_2 - \tfrac{2}{3}\nu &= 1 & 2 & 3 & 3 & 2 & 1 & 1 & 1 \\ \tfrac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \tfrac{1}{2}\mu_2 - \tfrac{2}{3}\nu &= 1 & 2 & 3 & 3 & 2 & 1 & 0 & 1 \\ \tfrac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \tfrac{1}{2}\mu_2 - \tfrac{2}{3}\nu &= 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ \tfrac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \tfrac{1}{2}\mu_2 + \tfrac{2}{3}\nu &= 1 & 2 & 3 & 2 & 2 & 2 & 1 & 2 \\ \tfrac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \tfrac{1}{2}\mu_2 + \tfrac{2}{3}\nu &= 1 & 2 & 3 & 2 & 2 & 1 & 1 & 2 \\ \tfrac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \tfrac{1}{2}\mu_2 - \tfrac{2}{3}\nu &= 1 & 2 & 3 & 2 & 2 & 1 & 0 & 2 \\ \tfrac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) - \tfrac{1}{2}\mu_2 + \tfrac{2}{3}\nu &= 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ \tfrac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \tfrac{1}{2}\mu_3 - \tfrac{2}{3}\nu &= 1 & 2 & 3 & 3 & 3 & 2 & 1 & 1 \\ \tfrac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \tfrac{1}{2}\mu_3 - \tfrac{2}{3}\nu &= 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ \tfrac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \tfrac{1}{2}\mu_3 - \tfrac{2}{3}\nu &= 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ \tfrac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \tfrac{1}{2}\mu_3 - \tfrac{2}{3}\nu &= 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ \tfrac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \tfrac{1}{2}\mu_3 + \tfrac{2}{3}\nu &= 1 & 3 & 5 & 4 & 3 & 2 & 1 & 3 \\ \tfrac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \tfrac{1}{2}\mu_3 + \tfrac{2}{3}\nu &= 1 & 2 & 3 & 2 & 1 & 1 & 1 & 2 \\ \tfrac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) - \tfrac{1}{2}\mu_3 + \tfrac{2}{3}\nu &= 1 & 2 & 3 & 2 & 1 & 1 & 0 & 2 \\ \tfrac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \tfrac{1}{2}\mu_3 + \tfrac{2}{3}\nu &= 1 & 2 & 3 & 2 & 1 & 0 & 0 & 2 \end{aligned}$
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E01117 (was)	189	■ 0.108	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = \left(\left(X_1, Y_1 \right) + \left(X_2, Y_2 \right) + \left(X_3, Y_3 \right) \right) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) \\ (X_2, Y_1) & (X_2, Y_2) \\ (X_3, Y_1) & (X_2, Y_3) \\ X \vee \boldsymbol{Y} & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X}Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) \\ (X_2, Y_1) & (X_2, Y_2) \\ (X_3, Y_1) & (X_2, Y_3) \\ X \vee \boldsymbol{Y} & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X}Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
0902.0431_E01117 (now)	189	—	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \wedge \boldsymbol{Y} = \begin{pmatrix} X_2 \wedge Y_3 - Y_2 \wedge X_3 \\ X_3 \wedge Y_1 - Y_3 \wedge X_1 \\ X_1 \wedge Y_2 - Y_1 \wedge X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Unresolved

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E01187	200	0.183	$\begin{aligned} &= (\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{a}, *(\boldsymbol{b} \wedge \boldsymbol{x})) - (\boldsymbol{a}_1, \boldsymbol{w})(\boldsymbol{w} \wedge \boldsymbol{a}_2, *(\boldsymbol{b} \wedge \boldsymbol{x})) \\ &+ (\boldsymbol{a}_2, \boldsymbol{w})(\boldsymbol{w} \wedge \boldsymbol{a}_1, *(\boldsymbol{b} \wedge \boldsymbol{x})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \\ &= \frac{2}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) - (\boldsymbol{a}_1, \boldsymbol{w})(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_2, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5) \\ &\quad + (\boldsymbol{a}_2, \boldsymbol{w})(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_1, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5), \\ &((\boldsymbol{b} \times *(\boldsymbol{a} \wedge \boldsymbol{x}))\boldsymbol{v}, \boldsymbol{w}) \\ &= (\boldsymbol{x} \wedge \boldsymbol{a}, \boldsymbol{w} \wedge *(\boldsymbol{b} \wedge \boldsymbol{v})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \\ &= (\boldsymbol{v}, \boldsymbol{w})(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{v}) + (\boldsymbol{a}_1, \boldsymbol{w})(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_2, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5) \\ &\quad - (\boldsymbol{a}_2, \boldsymbol{w})(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_1, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}). \end{aligned}$	(not rendered)	
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured refined: basis beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Flagged, not acted on

[illegible]

220 more flagged rows below the band, stated as a count: 30 component, 190 weak; confidence high 127, mid 51, low 42, none 0